

Wearable EMG Sensor for Detecting Bicep Contraction

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EE499 - Topics in Wearable Computing

Today's agenda



Project Goal

The goal of this project was to detect bicep muscle contraction using an EMG sensor connected to an Arduino.

The system was designed to:

1

Read electrical activity from the bicep

2

Send the signal to an Arduino

3

Display the values on a computer

4

Show a difference between relaxed and flexed muscle states

Background

1

Electromyography, or EMG, measures electrical activity produced by muscles.

2

When a muscle contracts, more muscle fibers become active, which causes a change in the EMG signal.

3

For this project, the focus was on the biceps because it is easy to access, easy to flex, and easy to test with a wearable sensor.





Device Design

Main components:

- Arduino Uno
- EMG muscle sensor
- Three electrode pads
- Jumper wires
- Laptop for data display

Basic wiring:

+VS goes to red rail

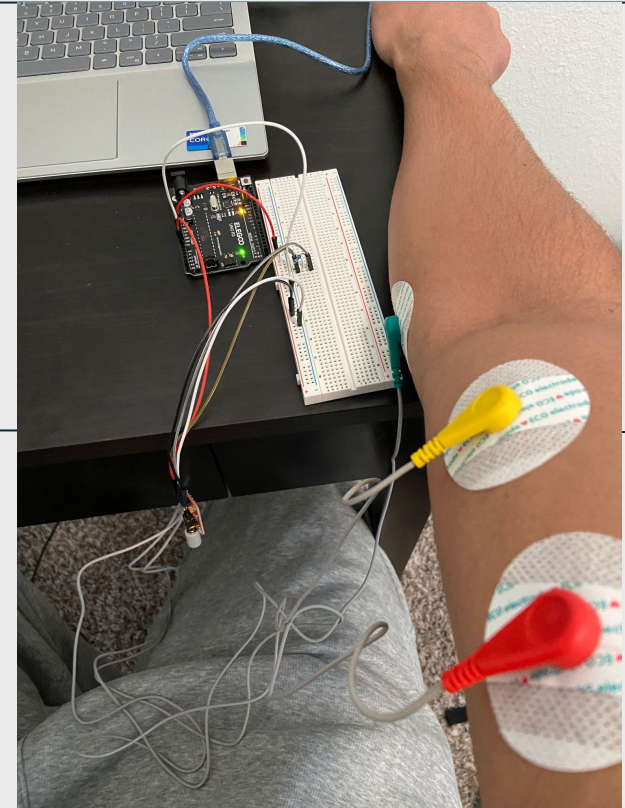
-VS goes to blue rail

GND goes to middle of resistors

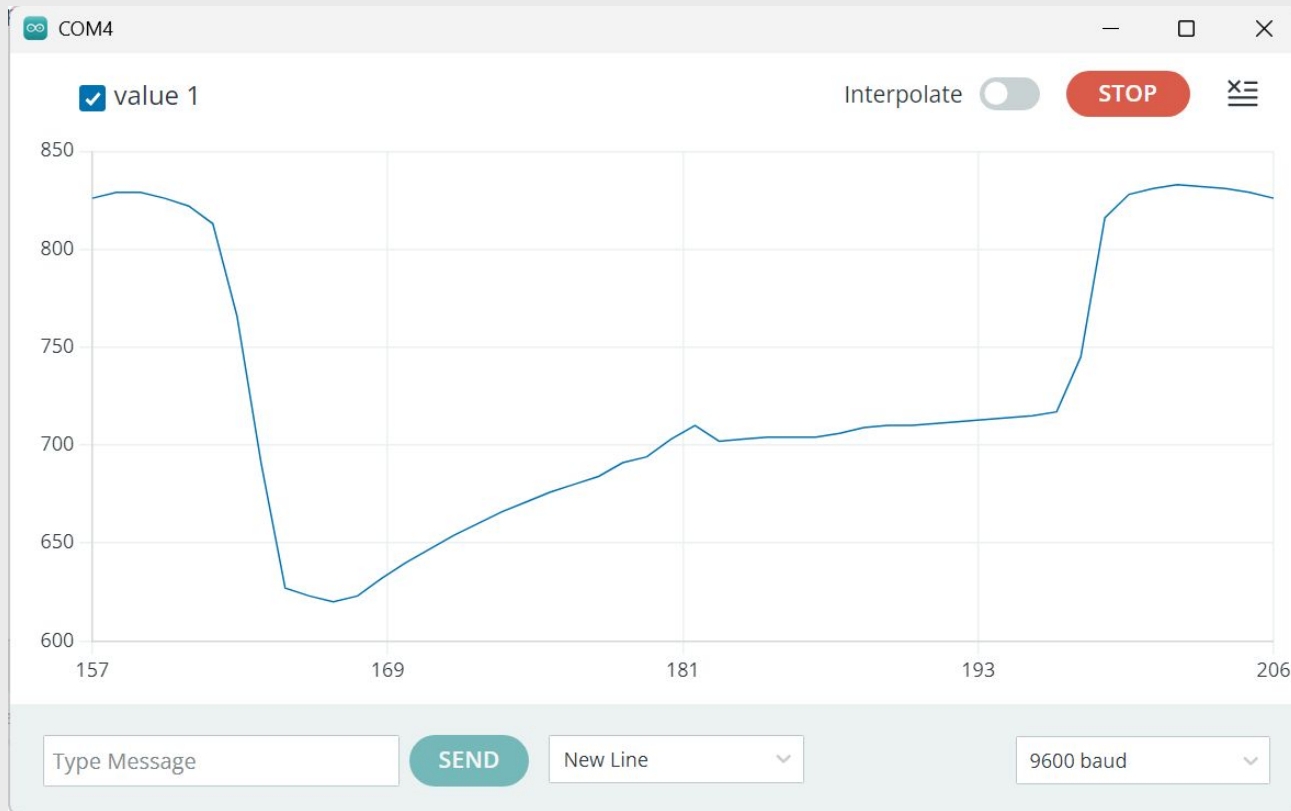
SIG goes to A0

Arduino 5V goes to red rail

Arduino GND goes to blue rail



Data Collection



The Arduino read the EMG signal through analog pin A0.

The signal was displayed as numbers using the Serial Monitor and also plotted.

The test followed this pattern:

- Relax arm
- Flex bicep
- Relax arm
- Flex bicep

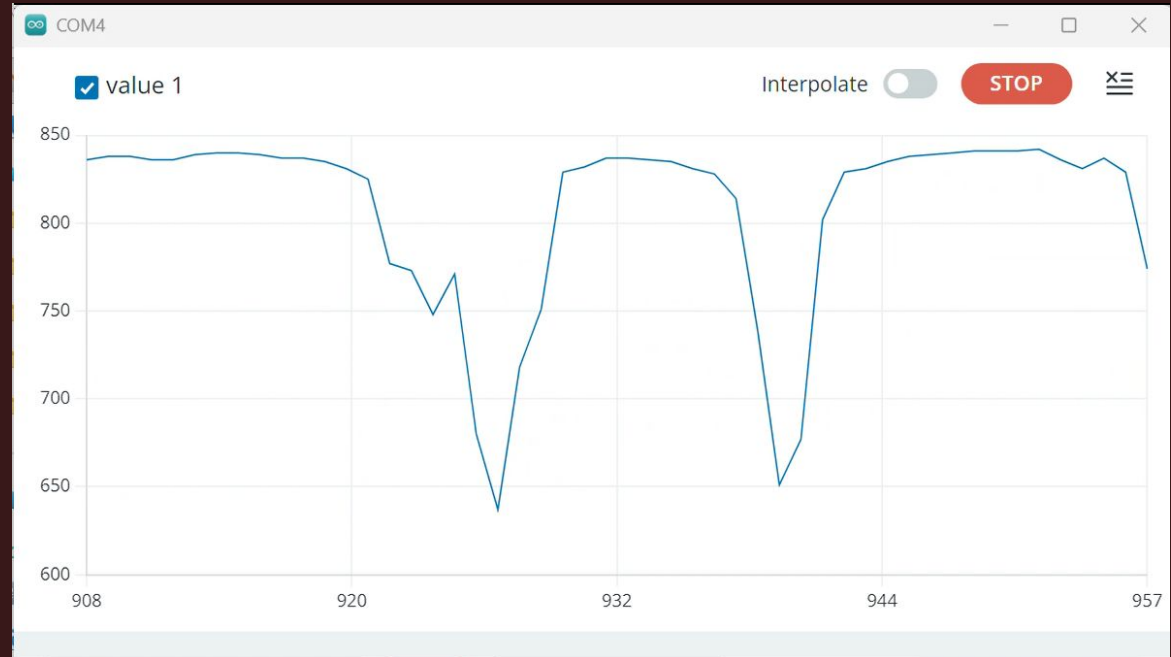
If value < 830, count as flexing

If value > 830, count as relaxed

Observations

Main observations:

- Relaxed muscle produced a more stable signal
- Flexing the bicep caused the signal to change
- The signal was noisy at times
- Electrode placement affected the quality of the readings





Issues

Some issues that came up during testing:

- The signal was noisy
- Electrode placement made a big difference
- The sensor was low-cost and not medical-grade
- The wiring setup required troubleshooting, especially with power to the EMG sensor
- The signal did not always behave perfectly

About
\$100

Conclusion



This project showed that a simple Arduino and EMG sensor setup can detect bicep muscle contraction.

The system was able to:

- Read muscle activity
- Display signal changes
- Show differences between relaxed and flexed states

Future Implementations/Ideas

Better EMG sensor



An actual wearable device



Types of curls



Other muscle groups



LED Indicator



App

Thank You